Udayaditya Palety

San Diego, CA | upalety@ucsd.edu | 530-608-6360 | linkedin.com/in/udayaditya-palety/

SKILLS

Technical: Machining, Laser Cutting, Soldering, 3D Printing (FDM, SLA, Polyjet), ISO Standardization, FMEA
Software: SolidWorks (CAD), MATLAB, Python, Arduino IDE (C++), DICOM, Netfabb, InVesalius, PreForm, LabView
Lab: Gel Electrophoresis, Restriction Digest, MicroPipetting, PCR, Cell Electroporation

EXPERIENCE

Systems Engineering Intern June 2026 – August 2026
Ultima Genomics
- Engineer a custom interferometry-based thin-film measurement system, including 3D-printed hardware and a Python GUI, collecting millions of wafer fluid-dynamics data points.
- Formulate a surfactant-stabilized coating solution and develop a custom spin-coating recipe for uniform, wash-removable coatings on 200mm silicon sequencing wafers.
- Conduct root-cause analysis on a new sequencing protocol using Ishikawa methodology, analyzing dozens of runs via an internal bioinformatics platform and AI agent.
- Design freeze-thaw stability experiments on reagent formulations, characterizing thaw kinetics and bleach efficacy to establish SOPs.

Process Engineer October 2023 – July 2025
Exela Pharma Sciences, LLC
- Designed and fabricated production equipment with SolidWorks to eliminate condensation issues in autoclaved syringe products, improving product quality and compliance.
- Created custom apparatus to improve OEE metrics in the aseptic filling suite, increasing efficiency and reducing downtime.
- Set up and validated pharmaceutical packaging and filling line equipment, developing SOPs and instructional documentation in compliance with cGMP guidelines.
- Maintained and optimized equipment in a cGMP manufacturing environment.

Undergraduate Researcher June 2022 – June 2023
UC Davis Cardiac Signaling Lab, Department of Pharmacology
- Performed translational research studying heart disease mechanisms for cardiotoxicity screening and drug R&D.
- Prototyped imaging apparatus using SolidWorks CAD and 3D printing to study cytotoxic drug effects on cardiomyocytes.
- Built an automated microscope assembly line using an Arduino microcontroller for high-throughput sample processing.

Vice President January 2020 – June 2023
B-Hours Engineering Club
- Led 60+ students across 15+ professional engineering projects as Vice President of the club.
- Directed an 8-member team and faculty partners to build an interactive parrot engagement device for the SF Zoo Aviary.
- Invented Mindex, a pocket-sized viral mitigation device that won first place in the BMES COVID-19 Design Competition.
- Organized 5+ technical workshops teaching soldering, CAD, and laser cutting to 50+ students.

PROJECTS

Digital Hand Dynamometer September 2022 – June 2023
Biomedical Engineering Senior Design Project
- Developed a digital hand dynamometer under Dr. Carolynn Patten to monitor and assess recovery in stroke patients.
- Machined 8+ aluminum components using an end mill and drill press; performed verification testing to ensure ISO compliance.
- Conducted clinical validation tests to demonstrate equivalence to the gold standard market product.
- Conducted user needs surveys and established Engineering Design Standards (EDS) and performance metrics.

UC Davis BMES Make-a-Thon Competition February 2022 – January 2023
- Awarded Most Creative Prototype and Most Promising Design in the 2022 and 2023 editions of the competition.
- Prototyped two medical devices with a team of four undergraduates for veterinary and NICU applications.
- Manufactured eight prototype components within 24 hours using laser cutting and rapid prototyping techniques.
- Presented final designs to a panel of six industry experts, highlighting design rationale, testing data, and user journey.

EDUCATION

University of California, San Diego
Master of Engineering in Bioengineering. *December 2026*
Relevant Coursework: Rapid Prototyping, Bioinstrumentation, Biotransport, Statistics, Biomedical Imaging, Circuits Lab
University of California, Davis
Bachelor of Science in Biomedical Engineering. GPA: 3.6. *June 2023*